



Substitution into equations and inequalities

Teacher resource

Question 1

$$2a - 1 = 3a + 2$$

Which value of a makes this equation true?

- ☐ $a = -3$
- ☐ $a = -1$
- ☐ $a = 1$
- ☐ $a = 3$

Answer: A

Skill: Students substitute values and solve an equation.

Additional questions

1. Is the equation true if $a = 0$?
2. How many values for a make this equation true?

Question 2

$$b = 7a^2$$

What is the value of b if $a = -3$?

- ☐ -63
- ☐ -42
- ☐ 63
- ☐ 441

Answer: C

Skill: Students substitute values and solve a simple quadratic equation.

Additional questions

1. What is the value of b if $7a = 28$?
2. What is the value of a if $b = 700$?
3. What are the values of b for $a = 5$ and $a = -5$?
4. What is the value of a if $b = 0$?

Question 3

If $x = 2$ and $y = 3$, choose the correct statement.

- ☐ $4x - 5y < -6$
- ☐ $4x - 5y = -6$
- ☐ $4x - 5y > -6$
- ☐ $-6 < 4x - 5y$
- ☐ $-6 = 4x - 5y$

Answer: A

Skill: Students evaluate inequalities.

Additional Questions

1. If $x = 3$ and $y = 2$, which statements are correct?
2. If $x = -3$ and $y = -2$, which statements are correct?

Question 4

Which pair of values makes the following statement true?

$$m < 4b - 3$$

- ☐ $m = 2$ and $b = 1$
- ☐ $m = 6$ and $b = 2$
- ☐ $m = 10$ and $b = 3$
- ☐ $m = 20$ and $b = 6$

Answer: D

Skill: Students substitute values in an equation.

Additional questions

1. If $m = 10$, what is the range of values of b ?
2. If $m = 0$, what is the range of values of b ?
3. If $b = 0$, what is the range of values of m ?

Student activity: Substitution into equations and inequalities

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